



University of Sri Jayewardenepura
Faculty of Applied Science
Department of Mathematics
BSc Degree First Year First Semester Course Unit Examination
2026 July/August

PMT 1022 Basics of Number Theory

Time allowed: 02 hours

Answer all questions

300 marks

- Structure of the paper:
Part A: Fifteen multiple choice questions ($15 \times 8 = 120$ marks)
Part B: Three Essay type questions ($3 \times 60 = 180$ marks)

PART A

- (1) From the following rational numbers, choose all rationals which have non-terminating decimal expansions.

$$\frac{1}{84}, \quad \frac{1}{250}, \quad \frac{1}{750}, \quad \frac{1}{1024}, \quad \frac{1}{256}$$

(a) $\frac{1}{84}, \frac{1}{750}$

(b) $\frac{1}{84}, \frac{1}{250}$

(c) $\frac{1}{84}, \frac{1}{750}, \frac{1}{750}$

(d) $\frac{1}{84}, \frac{1}{250}, \frac{1}{750}, \frac{1}{1024}$

(e) $\frac{1}{1024}, \frac{1}{250}, \frac{1}{256}$

(2) Let a, b are integers. What is true about followings?

- (i) If $a + b$ is divisible by 6 then at least one of a or b must be divisible by 6.
- (ii) If $a + b$ is d divisible by 6 then ab product must be divisible by 6.
- (iii) If ab is divisible by 6 then at least one of a or b must be divisible by 6.

- (a) All are false
- (b) Only (i) is true
- (c) Only (ii) is true
- (d) Only (iii) is true
- (e) Only (i) &(iii) are true

(3) What is true about following two sets?

$$A = \left\{ \frac{1}{3n+1} : n \in \mathbb{Z} \right\} \quad B = \left\{ \frac{1}{n} : n \in \mathbb{Z} \setminus \{0\} \right\}$$

- (a) The least element of A is -1 and the least element of B doesn't exist.
- (b) The least element of A is -1 and the least element of B is -1.
- (c) The least element of A doesn't exist and the least element of B doesn't exist.
- (d) None of the above answers is true.

(4) What is the largest power of 5 which exactly divides 125, 000, 000, 000?

- (a) 9
- (b) 10
- (c) 11
- (d) 12
- (e) 13

(5) Let $A = \{2^p 3^q : p, q \in \mathbb{Z}\}$. What is true about followings?

- (i) A has infinitely many odd integers.
- (ii) A has a least element.
- (iii) 324 is an element of A .

- (a) Only (iii)
- (b) Only (ii)
- (c) Only (ii) & (iii)

- (d) All are true
- (e) Only (i) & (iii)

(6) What are the last digits of 29^{18} and 7^{32} ?

- (a) 1 and 7 respectively
- (b) 9 and 1 respectively
- (c) 1 and 3 respectively
- (d) 1 and 9 respectively
- (e) None of the above

(7) Which is true about followings?

- (i) For every integer a , 3 is not a factor of $a^2 + 29$
- (ii) For every integer a , 3 is not a factor of $a^2 + 15$
- (iii) For every integer a , 3 is not a factor of $a^2 + 25$

- (a) All are false
- (b) Only (i) is true
- (c) Only (ii) is true
- (d) Only (iii) is true
- (e) Only (i) & (iii) is true

(8) Which is true about following?

- (i) Every subset of negative integers has a least element.
- (ii) The set of integers has a least element.
- (iii) Every subset of natural numbers has a least element.

- (a) Only (i)
- (b) Only (ii)
- (c) Only (iii)
- (d) Only (ii) & (iii)
- (e) Only (i) & (iii)

(9) Which is true about following?

- (i) If integers a and b relatively prime then at least one of a and b must be prime.
- (ii) If integers a and b both prime then a and b relatively prime.

- (a) Only (i) is true
- (b) Only (ii) is true
- (c) Both (i) & (ii) are true
- (d) Both (i) and (ii) False

(10) Which is true about following?

- (i) For any $k \in \mathbb{Z}$, $\gcd(5k + 4, 6k + 5) = 1$
- (ii) For any $k \in \mathbb{Z}$, $\gcd(5k + 3, 6k + 1) = 1$

- (e) Only (i) is true
- (f) Only (ii) is true
- (g) Both (i) & (ii) are true
- (h) Both (i) and (ii) are false

(11) Let a be any integer. What are all possible remainders when a^4 is divided by 5

- (i) 1
- (ii) 0, 1
- (iii) 0, 1, 2, 3, 4
- (iv) 0, 1, 4
- (v) None of the above

(12) Which is true about following?

- (i) For any integers a and b , 4 is not a factor of $a^2 + b^2 + 23$
- (ii) For any integers a and b , 4 is not a factor of $a^2 + b^2 + 16$
- (iii) For any integers a and b , 4 is not a factor of $a^2 + b^2 + 29$

- (a) All are false
- (b) Only (i) is true
- (c) Only (ii) is true
- (d) Only (iii) is true
- (e) Only (i) & (iii) is true

(13) Let p_n be the n^{th} prime number according to its magnitude. (I.e., $p_1 = 2$, $p_2 = 3$, $p_3 = 5$...etc. Which is true about following?

- (i) $p_1 p_2 p_3 \dots p_n + 1$ is prime for all $n \in \mathbb{N}$
- (ii) $p_n \equiv 1 \pmod{6}$ or $p_n \equiv 5 \pmod{6}$ for all $n > 2$ and $n \in \mathbb{N}$.

(14) If $a \equiv 0 \pmod{n}$ and If $b \equiv 0 \pmod{n}$ then, which is true about following?

- (i) $ab \equiv 0 \pmod{n^2}$
- (ii) $a^2 b^2 \equiv 0 \pmod{n^4}$
- (i) Only (i) is true
- (j) Only (ii) is true
- (k) Both (i) & (ii) are true
- (l) Both (i) and (ii) False

(15) What is the value of $2^{40} \pmod{13}$?

- (a) 3
- (b) 5
- (c) 7
- (d) 9
- (e) 1

PART B

Question 01

Prove or disprove each of the following universal statements.

- (i) Sum of any two positive irrational numbers is irrational.
- (ii) For a, b and c non-zero integers, if $a|bc$ then $a|b$ or $a|c$,
- (iii) For a, b non-zero integers, if $\gcd(a, b) = 1$, then $\gcd(a, -b) = 1$.
- (iv) For a, b, d integers, if $d|(a - b)$ and $d|(a + b)$, then $d^2|(a^2 - b^2)$.
- (v) For a, b integers, $a^2 + b^2 + 1$ is never divisible by 4.

(5 × 12 = 60 marks)

Question 02

- (a) Use the Euclidean Algorithm to obtain integers x and y satisfying:

$$\gcd(119, 272) = 119x + 272y.$$

(25 points)

- (b) Determine all integer solutions of the Diophantine equation $12x + 8y = 100$.

Find all positive integer solutions of above equation. {I.e., solutions (x, y) such that $x > 0$ and $y > 0$ }

(25 points)

- (c) Let p be a prime number and $p > 10$ write down all possible remainders if p is divided by 10.

Hence or otherwise prove that if p is prime and $p > 10$ then the last digit of p^4 is 1.

(10 points)

Question 03

Recall the theorem; “For a and b be integers not both zero, there exist integers x and y such that $\gcd(x, y) = ax + by$ ”

- (a) Using this theorem, prove that two integers a and b are relatively prime if and only if there exist integers x and y such that $ax + by = 1$.

- (b) Let a, b and c be integers. Prove that, If $a|c$ and $b|c$, with $\gcd(a, b) = 1$ then $ab|c$

- (c) Let a, b and c be integers. Prove that, if $a|bc$ with $\gcd(a, b) = 1$ then $a|c$.

- (d) Let p be a prime number and a, b are integers. Prove that if $p|ab$ then $p|a$ or $p|b$.

- (e) Prove that $\sqrt{p}$ is irrational for any prime p .

(5 × 12 = 60 marks)
